GrooveGen — where we are and what's next

A short update on the groove-music generation project · Daniel Llamas (Brain Music Lab / ATLAS, CU Boulder) · June 3, 2026

Hi Sophia — here's a plain-language summary of what the groove-generation system now does, how your dataset shaped it, what it already proves without any new participants, and the plan for the summer and fall. Nothing here requires action from you; it's a status share so we're aligned on direction.

The short version

I built GrooveGen: a small software system that generates short, looping pieces of music designed to make people want to move (the "groove" feeling). The key idea is a clean experimental separation. There is a constant musical bed — a fixed bass-and-chords loop that never changes — and on top of it sits a drum kit whose groove properties we can dial up and down with a handful of "knobs." Because the bed is frozen and only the drums move, any change a listener feels can be attributed to the knob we turned, not to the music drifting underneath.

How your dataset shaped the music

The constant bed is not arbitrary — its musical character is grounded in a secondary analysis of your dataset (122-participant survey, 643 unique tracks with audio features). I looked at the tracks participants chose for sensory-motor appreciation that also scored highest on danceability (n = 83), and matched the bed to their average profile: 120 BPM, 4/4, bass-forward, with prominent sustained harmony and a slightly positive mood. That became a bass-forward A-Dorian vamp (Am7 - D/A).

Two findings from your data justify the design

- Percussiveness and bass are the strongest danceability predictors in real recordings (standardized effects of +4.66 and +2.97, both clearly above zero). This is why the generator deliberately emphasizes drums and low-end.
- The 110-130 BPM tempo sweet-spot replicates: tracks in that band are about 9 danceability points higher ($p < .001$) — matching the literature and our tempo knob.
- The classic syncopation "inverted-U" does NOT appear in this naturalistic commercial audio. That sounds like a problem but it's actually the cleanest motivation for the whole project: the syncopation effect can't be recovered from passively-listened commercial songs, so it has to be studied with controlled, parametric stimuli — which is exactly what GrooveGen produces.

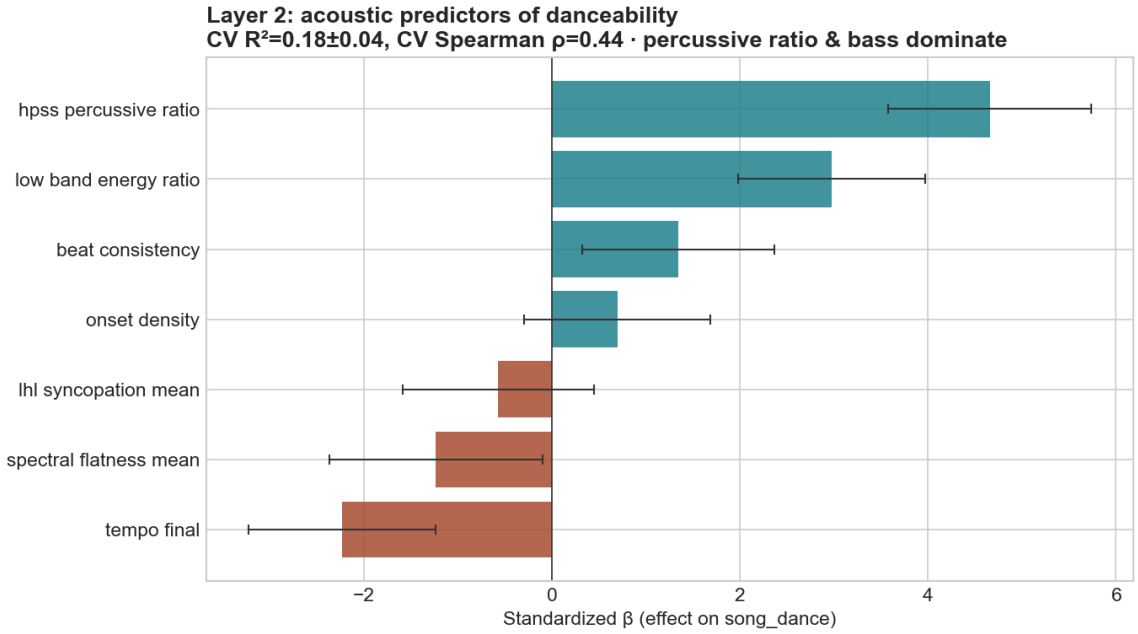


Figure 1. From your dataset: which acoustic features predict danceability. Percussiveness (hpss percussive ratio) and bass (low band energy ratio) dominate — the two dimensions the generator emphasizes.

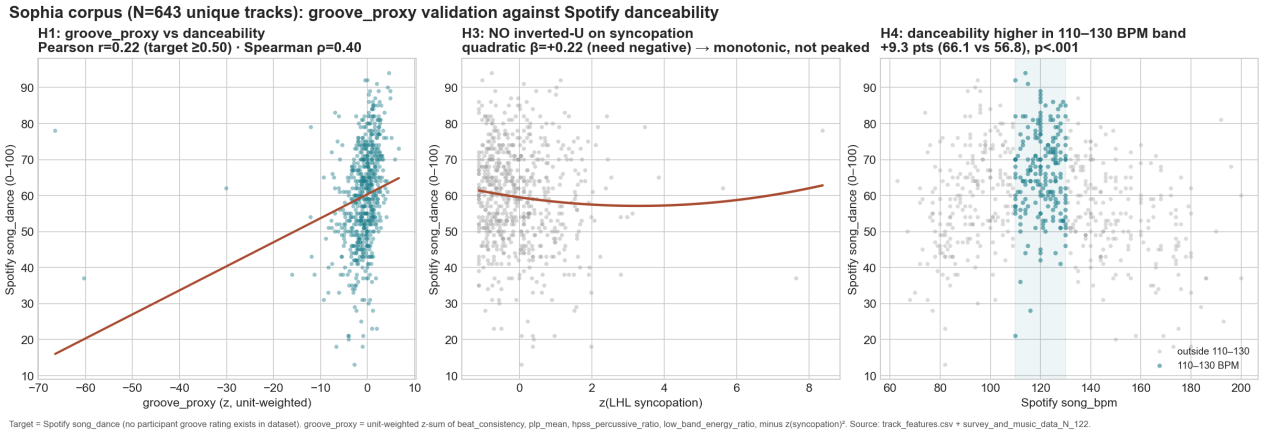


Figure 2. Three checks against Spotify danceability. The tempo sweet-spot replicates (right panel); the syncopation inverted-U does not appear in commercial audio (middle panel) — motivating controlled stimuli.

What the system can control (the knobs)

Each knob corresponds to a musical property the groove literature says matters. For each one I set four discrete levels spanning the published response curve, then verified by automated measurement that the rendered music actually hits those values — not just that we asked for them. All five knobs are now locked and validated:

- Syncopation (off-beat placement): 0 / 16 / 32 / 49 on the Witek index — level 2 sits right at the literature's groove sweet-spot, level 3 deliberately past it.
- Event density (how busy the drums are): 5 / 9 / 15 / 25 onsets per bar.
- Swing (the shuffle feel): 50% / 56% / 62% / 67%, straddling the 60-67% sweet zone.
- Tempo: 70 / 95 / 115 / 150 BPM (optional knob).
- Microtiming (timing looseness, used as a control): 0 to ~30 ms.

Crucially, the knobs move independently — turning syncopation up doesn't secretly change density — so a listener effect can be cleanly attributed to one variable.

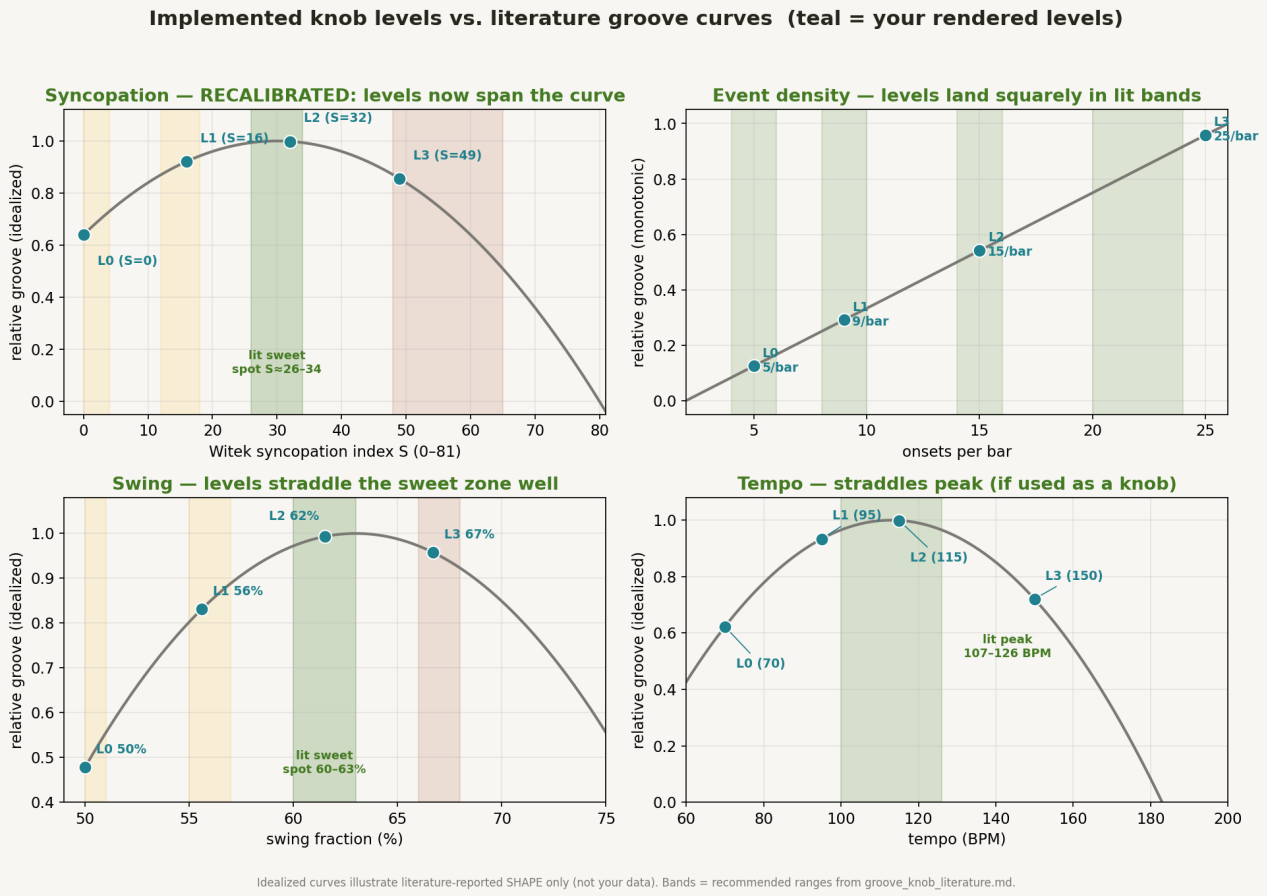


Figure 3. The teal dots are GrooveGen's actual rendered levels; the curves and shaded bands are the literature's groove response. Our levels now span each curve, including the syncopation sweet-spot.

What this already proves — with no new participants

Even before any human study, the system stands on three verified results: (1) every knob lands in its literature target band; (2) the knobs are independent of one another; and (3) the bed is truly constant across all stimuli. Together these make GrooveGen a reproducible, validated groove-stimulus generator — publishable on its engineering merits at a venue like NIME, SMC, or ICMC.

The plan — summer and fall

Because recruiting 40+ participants is realistically a fall activity, we're splitting the work so something is publishable sooner without waiting on the full study:

Summer (no large study required)

- Add a deep-learning bed generator. Instead of hand-authoring the bed, use a pretrained AI music model to generate it (starting with a symbolic/MIDI model so we keep full control of the drums), then overlay GrooveGen's knob-controlled drums. This adds a genuine "controllable-generation" novelty on top of the validated engine, and it's the realization of the "generate the bed, add our drums" idea.

- Write the system/methods paper describing the generator and its validation.
- Possible small pilot (n ~ 5). If a handful of listeners are available, run a quick feasibility check to confirm people perceive the knob levels as intended and that the rating questions work — this de-risks the full study (pending IRB).

Fall

- The full groove-rating study (n >= 40), using the validated 20-stimulus set, with self-report ratings first and PPG + respiration as a later stage.

Bottom line

The generation system is built and validated. Your dataset is what grounds the music and gives the study its scientific motivation. We can publish the system this summer and run the full human study in the fall — no time is lost by deferring recruitment. Happy to walk through any of this whenever useful.